

NETWORK DEVICES — ASSIGNMENT

Comprehensive Task Solutions and Network Architecture Analysis

Task 1: Simple Home Network Diagram & Architecture

Below is a structured architectural layout and description of a simple home network, detailing how the core networking hardware interconnects with end-user devices.

Network Topology Overview

In a standard modern household network, incoming broadband internet enters the premises via a Fiber-Optic connection or Cable Modem, which plugs directly into the primary Gateway / Router. The router serves as the central traffic controller, assigning local IP addresses via DHCP and translating private local addresses to public internet addresses via NAT (Network Address Translation).



Component Connection Breakdown

1. Router: Connects directly to the Internet Service Provider (ISP) broadband line. It establishes the local subnet, provides Wi-Fi access points, and manages firewall security.
2. Network Switch: Connected to one of the router's LAN ports via an Ethernet (RJ-45 Cat6) cable. The switch expands the number of available wired ports in a specific entertainment or work zone.
3. Smart TV: Connected to the network via an Ethernet cable plugged directly into the physical switch, ensuring high-bandwidth, buffer-free 4K video streaming.
4. Laptop / Workstation: Connected via an Ethernet cable plugged into the switch for ultra-low latency file transfers and coding work.
5. Smartphone: Connected wirelessly via the Wi-Fi broadcast generated by the router, enabling mobility throughout the home.

Task 2: Network Devices Comparison Table

The table below outlines four foundational network devices, comparing their core operational layer, function, and a daily real-world use case referencing common applications.

Network Device	OSI Layer	Primary Function	Real-World Use Case & Daily Services
Router	Layer 3 (Network)	Connects distinct networks together, routes packets across the internet, and assigns local IPs via DHCP.	Connecting home network to the WAN to stream 4K videos on YouTube or browse web pages across global servers.
Switch	Layer 2 (Data Link)	Connects multiple devices within a single LAN, forwarding frames based on MAC addresses.	Connecting a desktop PC, NAS drive, and smart TV in a home media room for high-speed local file sharing and LAN gaming.
Hub	Layer 1 (Physical)	Broadcasts incoming electrical signals across all ports indiscriminately without traffic filtering or destination awareness.	Legacy or educational test environments where basic signal repetition is needed (obsolete in modern high-speed networks).
Access Point (AP)	Layer 2 (Data Link)	Bridges wireless devices to an existing wired Ethernet network, expanding Wi-Fi coverage range.	Providing seamless Wi-Fi coverage across multiple floors in a multi-story home or office building for mobile devices and laptops.

Task 3: Data Journey Scenario — Ordering Food on Zomato

To understand how networking hardware functions in daily life, let's trace the exact journey of a data packet when you open the Zomato app on your smartphone, browse local restaurant menus, and place an order for delivery.

Step-by-Step Packet Journey

1. Initiation at the Smartphone: You tap on a restaurant listing and select your favorite pasta dish on the Zomato app. The smartphone packages this HTTP/HTTPS API request into digital data packets, adding your phone's Wi-Fi MAC address and local IP address.
2. Wireless Transmission via Access Point (AP): The phone emits radio frequency waves carrying the data packets. The Access Point (which may be a standalone unit or built directly

into your home Wi-Fi router) captures these radio waves, demodulates them into electrical signals, and injects them into the wired Ethernet infrastructure.

3. Local Switching via Network Switch: If your home network utilizes an Ethernet switch connected between various rooms and the central router, the electrical frames pass through the switch. The switch inspects the destination MAC address to ensure the packets are correctly forwarded to the router's LAN port without unnecessary flooding.

4. Routing and WAN Translation via the Router: The packets arrive at the home Router. The router realizes the destination IP address (Zomato's cloud server hosted on AWS) is outside the local network. It performs Network Address Translation (NAT), replacing your private local IP with your ISP public IP address, and encapsulates the packet into an internet-routable format.

5. Internet Transit and Cloud Delivery: The router transmits the packet via fiber optic or cable broadband to your Internet Service Provider (ISP). The ISP's core routers guide the packet across fiber backbone networks through internet exchange points (IXPs) until it reaches Zomato's cloud servers. Zomato processes the order, confirms payment, and sends a confirmation response back along the exact reverse path to light up your phone screen with 'Order Placed Successfully!'

Task 4: Connecting a Distant Gaming PC via Ethernet

Situation: You have a standard wireless Wi-Fi router located in the living room, but your high-performance gaming PC is located in a distant bedroom. The PC only has an RJ-45 Ethernet port (no Wi-Fi card installed) and cannot easily run a long physical Ethernet cable across the house due to distance and doorways.

Proposed Solution 1: Powerline Ethernet Adapter Kit

How it works: A Powerline adapter kit consists of two modular units that utilize your home's existing electrical wiring to transmit network data.

- Setup: Plug the first Powerline adapter into an electrical wall outlet right next to your Wi-Fi router and connect them using a short Ethernet cable. Plug the second Powerline adapter into an electrical outlet near your gaming PC in the bedroom, and run an Ethernet cable from that adapter directly into your PC's LAN port.
- Mechanism: The adapter near the router modulates Ethernet data signals into high-frequency electrical signals that travel over the home's AC power circuits. The receiving adapter demodulates the signal back into standard Ethernet frames. This provides a remarkably stable, low-latency wired connection ideal for competitive gaming without drilling holes or running long cables.

Proposed Solution 2: Wireless Bridge / Wi-Fi Extender with Ethernet Port

How it works: A Wireless Range Extender or dedicated Client Bridge device equipped with an Ethernet port acts as a wireless-to-wired converter.

- Setup: Place the wireless bridge halfway between the living room router and the bedroom gaming PC (ensuring it remains within strong Wi-Fi reception range). Plug an Ethernet cable from the bridge's LAN port directly into the gaming PC's Ethernet port.
- Mechanism: The wireless bridge connects wirelessly to your main router's Wi-Fi network acting as a client station. It receives Wi-Fi radio transmissions from the router and bridges them directly into the physical Ethernet cable plugged into your PC. This allows a device lacking wireless hardware to seamlessly tap into your existing wireless network infrastructure.